

WFM Agent Performance Analysis

Python, SQL, and Power BI Project



Project Overview



Goal

Identify productivity patterns and develop Productivity Score model



Metrics

AHT, Occupancy, Utilization, Shrinkage



Workflow

Python → SQL → Power BI

Tools and Technologies



Python

Pandas for data analysis and feature engineering



MySQL

Data transformation, KPI calculations, views, ranking logic



Power BI

Interactive dashboards and operational performance visualization



Dataset Overview

Agent Information

- Agent ID, Name, Team
- Gender, Age, Shift Timing
- Date, Attendance Status

Performance Metrics

- Calls Handled
- Average Talk Time
- Break Time, Training Time
- Login Time, Hold Time

Python Data Analysis

01

Data Cleaning

Convert dates to datetime, handle missing values, standardize time fields

03

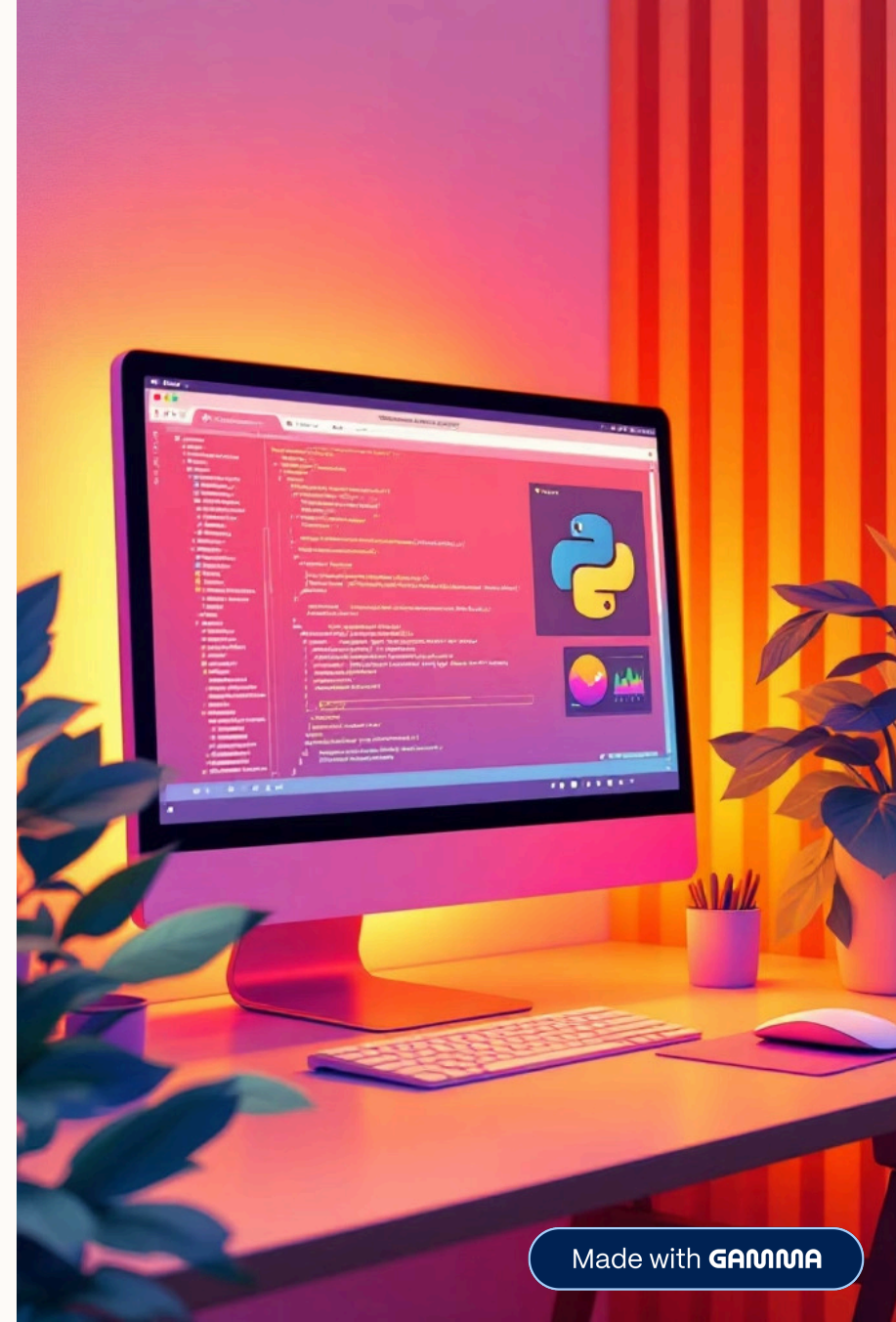
Data Export

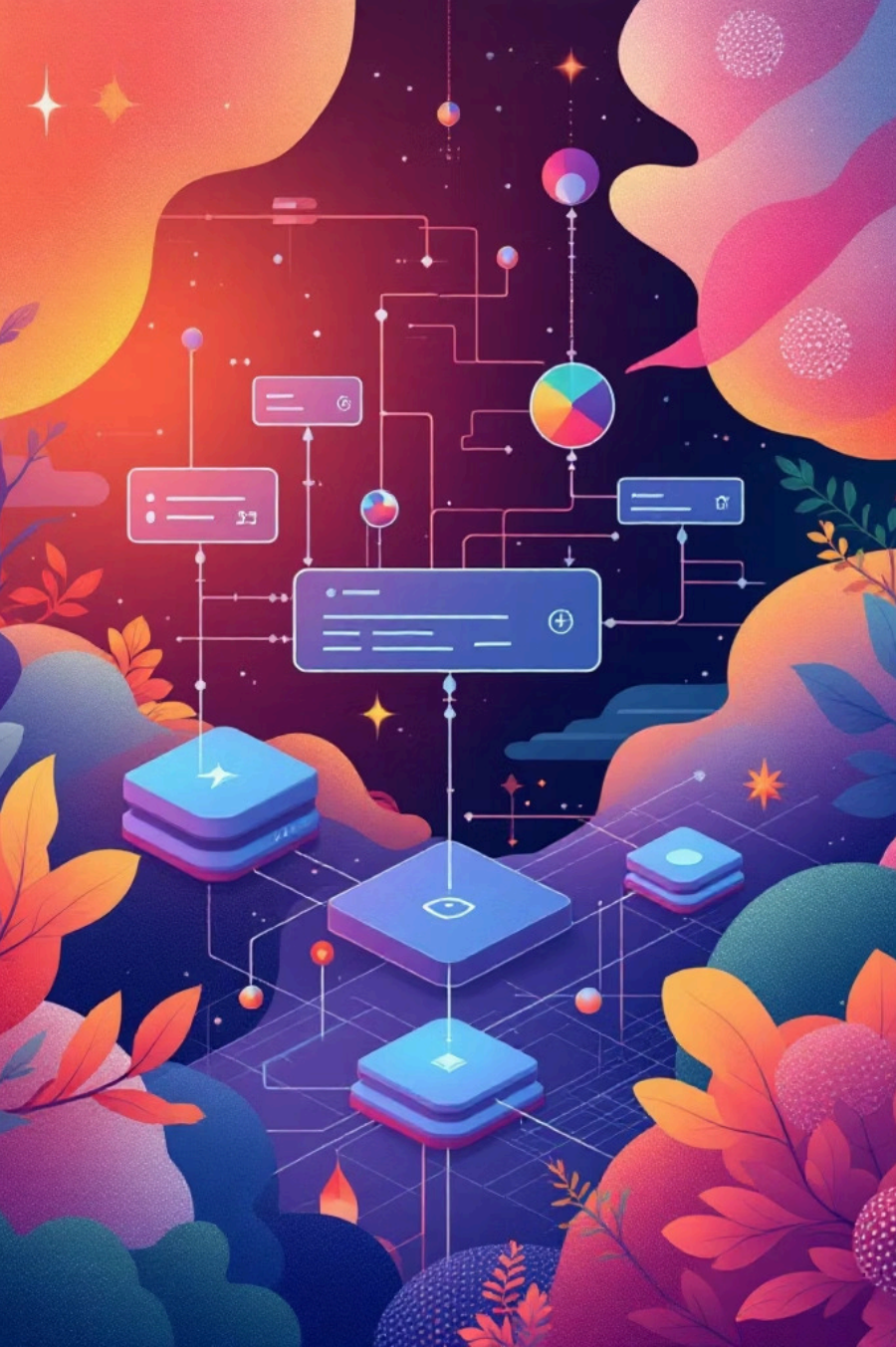
Load cleaned data to MySQL database

02

Feature Engineering

Create calculated metrics and derived columns





SQL Data Transformation



Window Functions

Rank agents within teams



Views

Create structured analytical views



Aggregations

Calculate KPIs and metrics

Workforce Metrics

1

Total Handle Time

$(\text{Talk Time} + \text{Hold Time}) \times \text{Calls Handled}$

2

Average Handle Time

$\text{Total Handle Time} / \text{Calls Handled}$

3

Effective Work Hours

$\text{Total Login} - \text{Break} - \text{Training}$

4

Occupancy

$\text{Total Handle} / \text{Effective Work}$

5

Utilization

$\text{Effective Work} / \text{Total Login}$

6

Shrinkage

$(\text{Break} + \text{Training}) / \text{Total Login}$

Productivity Scoring Model



Productivity Score

$\text{AHT Score} \times 0.4 - \text{Occupancy} \times 0.3 - \text{Utilization} \times 0.3$



Ranking

SQL window functions identify top performers

Power BI Dashboard

Executive Overview

Total Calls, AHT, Occupancy, Utilization

Team Performance

Team Occupancy, Calls by Age, Distribution

Interactive Filters

Team, Gender, Age Category



Key Insights



Team Performance

Certain teams handle higher call volumes consistently



Shift Analysis

11 AM – 8 PM shift handles highest call volume



Age Factor

Agents aged 21–27 show higher productivity levels



Utilization Impact

Strong correlation with overall productivity score



Shrinkage Variation

Significant differences across teams and periods